pIAno: Symbolic Music Generation from ABC Notation and Images

Antoine Delacour Eloïse Trabut
École Polytechnique
{antoine.delacour, eloise.trabut}@polytechnique.edu

March 31, 2026

Abstract

*We present **pIAno**, a symbolic music generation system based on ABC notation that extends standard language-model approaches to a multimodal setting. In a first stage, a GPT-style Transformer is trained on BPE-tokenized sequences from 50,000 tunes drawn from the Irish Massive ABC Notation Dataset (∼216,000 tunes), reaching a validation perplexity of 7.4. A memorization analysis confirms that the model recombines learned patterns rather than copying training data (maximum sequence similarity: 0.46; longest copied fragment: 14 characters). In a second stage, we introduce an image-conditioned generation pipeline that requires no paired image–music dataset. CLIP embeddings of natural-language tune descriptions serve as surrogate images during training, so that real photographs can directly condition generation at inference time. Mistral-7B is fine-tuned with 4-bit quantization and LoRA (<1% of parameters updated) on a balanced dataset of* $2\times2{,}011 = 4{,}022$ *tunes. Qualitative evaluation confirms that fine-tuning is necessary for syntactically valid ABC output. A generation example shows that the system can associate the melancholic mood of sunset photographs with a dark musical mode without any explicit musical instruction, representing a promising first step toward cross-modal music generation; occasional mood mismatches remain and are discussed as directions for future work.*

1. Introduction

Music generation can be approached at different levels of representation, ranging from raw audio signals to symbolic musical descriptions. This project focuses on **symbolic music generation**, where music is represented as a sequence of discrete symbols.

Our work relies on **ABC notation**, a text-based format for symbolic music, and explores how language-modeling methods can be adapted to generate musical sequences. Because ABC notation is written as plain text, it is particularly well suited to autoregressive Transformer models, which can be trained to predict the next token in a sequence.

The project combines two complementary perspectives. First, it studies symbolic music generation from ABC sequences and investigates the role of tokenization strategies in musical modeling. Second, it explores a multimodal direction in which image representations are used to condition the generation of musical sequences via CLIP [5] and a fine-tuned Mistral-7B [3] model.

2. ABC Notation and Dataset

ABC notation is a text-based representation for symbolic musical scores [4]. It encodes musical information using ASCII characters and was originally designed to make folk melodies easy to write, store, and share in plain text. A typical ABC tune is composed of two main parts: a **header**, which contains musical metadata, and a **body**, which describes the sequence of notes. The most common header fields are: `X:` (reference number), `T:` (title), `M:` (time signature), `L:` (default note length), and `K:` (key signature).

Example of ABC Notation Score

```
X:1
M:4/4
L:1/8
K:G
G2 | G2 G>B B>G | G>AB>G G>B |
```

ABC notation is particularly convenient for machine learning because it represents music directly as text, making it possible to use standard language-modeling techniques such as autoregressive Transformers without requiring a complex intermediate representation.

The experiments rely on the **Irish Massive ABC Notation Dataset** [1], a large collection of traditional Irish folk melodies encoded in ABC notation. The dataset contains approximately **216,000 tunes**, the vast majority of which are **monophonic melodies**. This makes it particularly well suited for sequence modeling, as its musical structure is

simpler than that of highly polyphonic corpora.

Since the models considered here generate music token by token and do not explicitly capture harmony or long-range hierarchical structure, monophonic melodies are easier to learn and reproduce effectively.

3. BPE-Based Transformer for Symbolic Music

3.1. Tokenization

Before training a generative model, symbolic musical sequences must be converted into discrete tokens. This step is essential because tokenization directly shapes the way a model processes music: it affects the length of input sequences, the computational cost of training and inference, and the ability of the model to capture both local regularities and higher-level musical structure.

Rather than relying on a manually designed musical vocabulary, we apply **Byte Pair Encoding (BPE)** directly to the ABC text corpus. Starting from a character-level representation, the algorithm iteratively merges the most frequent symbol pairs, yielding a vocabulary of **1024 tokens**. Many learned tokens correspond to recurring musical fragments such as `BAG`, `dcA`, `|:`, or `:|`.

Compared to character-level tokenization, BPE reduces effective sequence length and allows the model to capture recurring musical motifs more efficiently. However, because the vocabulary is driven by statistical co-occurrence rather than musical semantics, some learned tokens may correspond to fragments that do not map directly to meaningful musical events. This reflects a general trade-off between **statistical compression** and **musical interpretability**, as discussed in models such as NotaGen [6] which rely on manually designed tokenization schemes instead.

3.2. Model and Training

Symbolic music generation is formulated as a sequence modeling task. Given a sequence of tokens $t_1, t_2, \ldots, t_n$, the model learns to estimate the conditional probability $P(t_{i+1} \mid t_1, \ldots, t_i)$, and generates new sequences by sampling autoregressively from this distribution.

The model is a **GPT-style Transformer** kept intentionally small in order to maintain a simple and interpretable pipeline: 4 layers, 4 attention heads, embedding dimension 256, context length 256, and vocabulary size 1024. Each token is first mapped to a learned embedding combined with positional embeddings, then processed through successive Transformer layers where self-attention captures dependencies across the context window.

The model is trained on up to **50,000 tunes** (90/10 train/-val split, AdamW optimizer, $\mathrm{lr} = 3 \times 10^{-4}$, 3000 steps) using a next-token cross-entropy objective. Both training and validation losses decrease steadily; at convergence the validation loss reaches $\approx$**2.0**, corresponding to a perplexity of **7.4**, suggesting that the model successfully captures local musical patterns.

3.3. Generation and Memorization Analysis

After training, the model generates new sequences by sampling tokens autoregressively. Generated outputs are written in ABC notation and can be converted into musical scores and audio using standard symbolic music tools. The generated score is shown in Figure 1 and an audio rendering is available online. In many cases, generated sequences follow valid ABC syntax and produce melodies that resemble Irish folk music. However, because the model generates tokens without explicitly modeling higher-level structure, longer generations may exhibit repetitive patterns or inconsistencies in musical phrasing.

Example of Generated ABC Notation

```
X:1
T:Generated Tune
M:4/4
L:1/8
K:G
G2 | G2 G>B B>G | G>AB>G G>B | d>ef>g | B3 G |
A>BA>G G>F | B>AB>A G>A | A>GA>B B>AB>G D |
E>GA>B G>B | G>B B>g G>G | A>AG>G G>B |
A>GA>B G>B | d>ed>e d>G G>B | A>BA>B F>G |
A>GA>B G>B | c>AB>g G2 | A>Bc>d B>d |
```



Figure 1. Example of a melody generated by the Transformer model. The output follows valid ABC syntax and exhibits rhythmic patterns characteristic of Irish folk music.

To evaluate memorization, generated melodies were compared against the training set using two metrics. First, the maximum similarity between generated sequences and dataset entries was **0.46**, indicating that generated melodies remain substantially different from training examples. Second, the longest exact substring shared with any training example was only **14 characters** (`K:G\nG2 | G2`), a very common pattern in ABC notation whose presence does not indicate memorization of complete tunes. Overall, the model recombines short learned musical patterns rather than reproducing entire training sequences.

4. Multimodal Extension: Music from Images

4.1. Motivation and Key Challenge

The goal of this extension is to condition music generation on one or several **images**: the user uploads photographs, and the system produces a melody whose mood and style reflect the visual content. To refine this cross-modal synthesis, the user can assign specific weights w_i to each image, defining their relative importance in the final composition.

Training an image-to-music system typically requires large-scale datasets of paired (image, ABC notation) examples. However, no such publicly available resource exists: while existing multimodal music datasets link audio or MIDI to textual descriptions, none bridge the visual domain and symbolic notation. We overcome this using **CLIP** [5] as a bridge between modalities. CLIP is a vision model trained to place images and their textual descriptions close together in the same 512-dimensional latent space. During training, musical pieces are described in natural language and encoded with CLIP's *text* encoder, producing a 512-dim vector that serves as a surrogate image. At inference, any real photograph can replace this vector, because a photo of a stormy sea and the sentence "a dark Irish tune in D minor" occupy the same neighbourhood of the latent space.

4.2. Model Choice

A small Transformer trained from scratch on folk melodies has no prior knowledge of language or the world, and would have no meaningful way to interpret a CLIP embedding. **Mistral-7B-v0.1** [3] is therefore selected for its strong sequence-modeling priors and its computational efficiency via Grouped-Query Attention (GQA) and Sliding Window Attention (SWA). Since ABC notation is purely textual, no structural modification to Mistral is required. The model is loaded in **4-bit NF4 quantization** (with double quantization) and adapted with **LoRA** [2], updating fewer than 1% of total parameters. Visual conditioning uses **CLIP ViT-B/32** (`laion2b_s34b_b79k`).

4.3. Dataset Construction

We select 10,000 tunes from the Irish Massive ABC Notation Dataset and process each into a training example through five steps. **(1)** Parse `K:`, `M:`, `L:`; discard incomplete tunes and exact duplicates. **(2)** Assign an emotional label from the musical mode: minor, Dorian, Phrygian, and Aeolian modes $\rightarrow$ *sad and dark*; major $\rightarrow$ *happy and upbeat*. **(3)** Build a natural-language prompt combining emotion, meter, base note duration, key, and structural control code, *e.g.*:

> *"A sad and dark traditional Irish tune in 4/4, base note 1/8, key Emin, structure:* `|:A|A|B|B:|`*"*

(4) Encode the prompt with CLIP ViT-B/32 in batches of 128 on GPU, producing a 512-dim L2-normalised surrogate image vector. **(5)** Balance the dataset by downsampling the majority class so that $n_{\text{sad}} = n_{\text{happy}}$ (observed: 2,011 each; total 4,022 tunes), preventing the model from collapsing to one mood.

Each example is stored as a JSONL record with two fields: `text` (Mistral instruction-following format including an `<IMAGE_EMBED>` placeholder, with the instruction drawn from 5 random phrasings to prevent prompt memorization) and `image_features` (the 512-dim CLIP vector).

4.4. MultimodalCollator

Standard collators handle only text and would silently ignore the `image_features` field. Our **MultimodalCollator** performs four operations per batch: **(1)** tokenize and left-pad sequences using EOS as padding token, producing the attention mask; **(2)** stack precomputed CLIP vectors into a $(\text{batch}, 512)$ float16 tensor; **(3)** compute token embeddings explicitly via `model.get_input_embeddings().cpu()`, yielding a $(\text{batch}, \text{seq}, 4096)$ tensor — the `.cpu()` call is critical to avoid premature GPU transfer; **(4)** locate `<IMAGE_EMBED>` in each sequence and overwrite the first 512 values of its 4096-dim embedding with the CLIP vector. The modified embedding matrix is returned directly to `SFTTrainer`, bypassing Mistral's internal embedding lookup entirely and preserving the visual injection.

4.5. Fine-Tuning with LoRA

Rather than retraining Mistral's 7 billion parameters from scratch, LoRA inserts small trainable matrices into specific layers while the original weights remain frozen, updating fewer than 1% of total parameters. The model is fine-tuned for **150 steps** using `SFTTrainer` from the TRL library. Key hyperparameters are listed in Table 1. Restricting LoRA to attention projections only yielded unsatisfactory results; covering MLP layers proved necessary, suggesting that these layers play a key role in encoding musical style. The training loss drops from $\sim$3.1 to $\sim$1.5 in the first 20 steps and stabilises around 1.0 from step 90 onward (Figure 2), suggesting convergence within the available compute budget.

4.6. Inference Pipeline

At inference, the system proceeds in five stages. **(1) Encoding**: each uploaded image is converted to RGB, preprocessed with CLIP's `preprocess` transform, and encoded via `clip_model.encode_image()`, then L2-normalised. **(2) Weighted fusion**: embeddings are combined as $\mathbf{e}_{\text{fusion}} = \sum_i w_i \,\text{CLIP}(\mathbf{I}_i) / \sum_i w_i$, cast to float16, and placed in a $(1, 1, 4096)$ tensor. **(3) Visual injection**: the fused embedding overwrites the `<IMAGE_EMBED>` position in the prompt embedding before autoregressive generation (`temperature`=0.75, `repetition_penalty`=1.2, `max_new_tokens`=350,

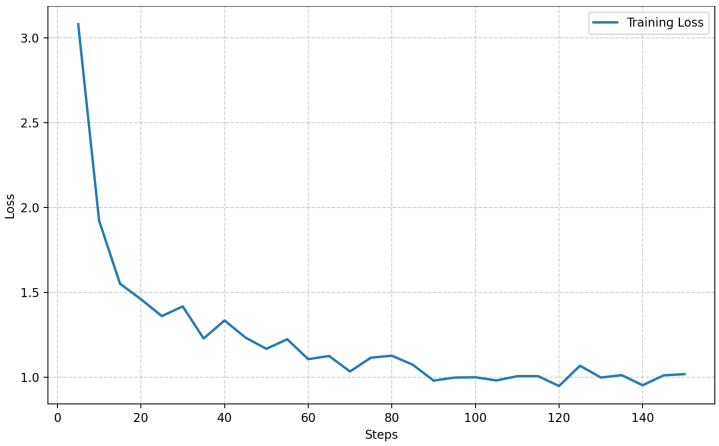


Figure 2. Training loss over the 150 LoRA fine-tuning steps. The loss drops sharply from ∼3.1 to ∼1.5 in the first 20 steps, then stabilises around 1.0 from step 90 onward.

Parameter	Value	Rationale
Quantisation	NF4 4-bit + dq	14 GB → ∼4 GB
LoRA r/α	16 / 32	α=2r, stable scaling
LoRA dropout	0.05	Light regularisation
Target modules	q,k,v,o,gate,up,down	Attn. + MLP both required
Optimiser	paged_adamw_8bit	Offloads Adam to CPU
Learning rate	10^{-4}, cosine	Higher → incoherent ABC
Warmup steps	20	Avoids instability at start
Eff. batch size	2×4=8	Gradient accumulation
Weight decay	0.01	ℓ_2 on LoRA params
Max seq. length	512 tokens	≈16 ABC measures

Table 1. LoRA fine-tuning hyperparameters.

`do_sample`=True; `do_sample`=False produces near-identical outputs regardless of image, defeating the purpose of conditioning). **(4) Cleaning**: a regex removes characters outside the valid ABC alphabet. **(5) Export**: the cleaned ABC is normalised by `nettoyer_abc` (which fills missing mandatory headers and reorders them as `X:`→`T:`→`M:`→`L:`→`K:`), then parsed by `music21` and rendered to WAV at 44,100 Hz by FluidSynth using the FluidR3_GM soundfont.

4.7. Generation Example and Evaluation

The two input images used for the generation example are shown in Figure 3: both are harbour-sunset photographs conveying a calm, melancholic mood. Image 1 (marina at sunset, w_1=0.7) and Image 2 (canal at dusk with warm orange and purple tones, w_2=1.3, given higher weight as its colours are darker and more intense) are fused through the inference pipeline. The generated melody is in **Dorian** mode, a minor-like scale commonly used in Celtic folk music to express sad or introspective feelings, consistent with the mood of the two images and without any explicit musical instruction being provided. The generated audio is available online.

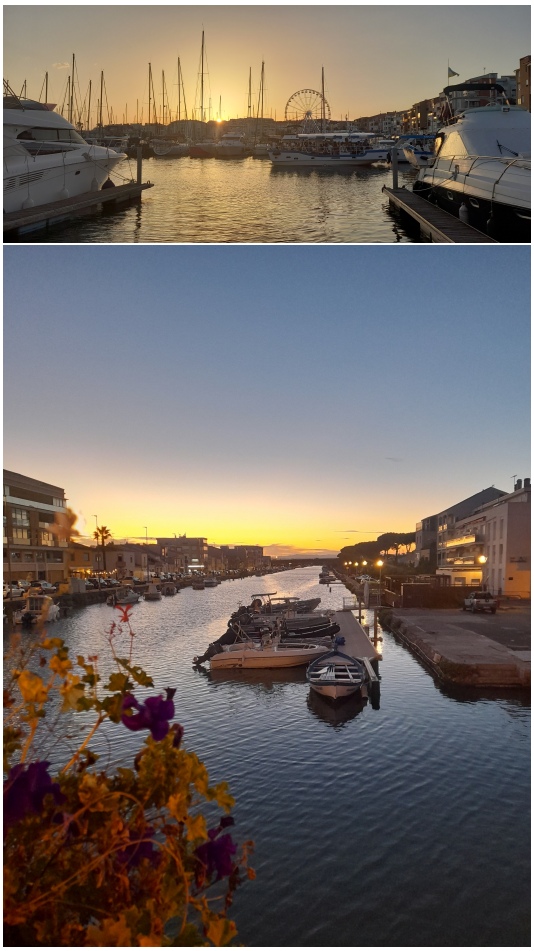


Figure 3. The two input images used for the generation example. **Top**: marina at sunset (w_1=0.7). **Bottom**: canal at dusk (w_2=1.3). Both convey a calm, melancholic atmosphere. The system generates a melody in Dorian mode without any explicit musical instruction. Generated audio available online.

To verify the contribution of fine-tuning, we run the same image through the model with LoRA active (**pIAno**) and disabled (**base Mistral**), with identical inputs and generation parameters (`temperature`=0.7, `repetition_penalty`=1.2, `top_p`=0.9, `top_k`=50, `max_new_tokens`=200, `do_sample`=True). The two outputs are shown in Figure 4.

The pIAno output is a single, well-formed Irish tune in E Dorian with a correct two-part structure and idiomatic ornaments (dotted rhythms, chord symbols). The base Mistral output is structurally incoherent: the title field `T:[1]` is malformed, the model generates a second tune header `X:2` mid-sequence, and includes a `w:o` field that has no meaning in ABC notation. This confirms that fine-tuning is

pIAno output (LoRA active)

```
X:1
L:1/8
M:4/4
K:Edor
|:"A" c>e dB, A,2 | Bd ef gcBA |"D7" AF GF
GABG | ABcA F3 ::
z"Cm" fgfa afba |"Am" gabag fedc |"F#m" fefg
afea |"Gb" bgae dcde ||
```

Base Mistral output (LoRA disabled)

```
X:1
T:[1]
M:6/8
L:1/4
K:Gmaj
|: e2g | a gf ecA AGA | B2B A2c dBG,E | G3 FED
EFG | AGF EDE DFA,G |
e2g afec cBA,G | AGE FAce dega,g | (fg)af edc
dcAG | (3Bcd egf ege^d ||
w:o
X:2
T:[2]
M:6/8
L:1/4
R:Jig
K:Gmaj
(~c2d2) B2 c ~A3 (afe) fged | (bac) ~B3 (agf)
egd ^edef |
ga ~b2 aga bgc | AGB ~A2 geg ^edef
```

Figure 4. Comparison: same image passed through the model with LoRA active (pIAno) and disabled (base Mistral), all generation parameters held identical.

necessary for the model to produce syntactically valid and well-structured ABC scores.

5. Conclusion

This project studied symbolic music generation with Transformers using ABC notation, and extended it to a multimodal setting where music is generated from images. In the first part, a BPE + Transformer pipeline was implemented and analyzed. The results show that this approach is able to learn local melodic regularities and generate plausible ABC sequences that stylistically resemble Irish folk music. The memorization analysis confirms that the model recombines learned patterns rather than reproducing training examples: highest similarity 0.46, longest copied fragment 14 characters. Tokenization emerges as a central design choice in symbolic music generation, affecting both computational efficiency and the type of musical structure the model can learn.

The second part introduced a multimodal extension in which image representations condition the generation of musical sequences. By using CLIP as a bridge between modalities, the system avoids the need for a paired image–music dataset. Mistral-7B is fine-tuned with LoRA on a balanced dataset of 4,022 tunes, updating fewer than 1% of total parameters. The comparison between fine-tuned and base model output confirms that fine-tuning is necessary for syntactically valid and well-structured ABC scores. The generation example shows that the system can associate the melancholic atmosphere of sunset photographs with a dark musical mode without any explicit musical instruction, representing a promising first step toward cross-modal music generation.

Despite some mismatch between the visual mood of the input image and the emotional character of the generated melody, the results are encouraging given the constraints: no paired image–music dataset, a limited number of training steps, and a general-purpose vision encoder. Several directions could be explored in future work: a larger and more emotionally diverse training dataset, a vision encoder more specifically trained on affective or artistic content, and an increased LoRA rank or fine-tuning budget to better capture subtle cross-modal correspondences.

References

[1] Irish massive abc notation (irishman) dataset. https://huggingface.co/datasets/sander-wood/irishman, 2023. Hugging Face dataset. 1

[2] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. 3

[3] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mistral 7b, 2023. 1, 3

[4] Xingwei Qu, Yuelin Bai, Yinghao Ma, Ziya Zhou, Ka Man Lo, Jiaheng Liu, Ruibin Yuan, Lejun Min, Xueling Liu, Tianyu Zhang, Xinrun Du, Shuyue Guo, Yiming Liang, Yizhi Li, Shangda Wu, Junting Zhou, Tianyu Zheng, Ziyang Ma, Fengze Han, Wei Xue, Gus Xia, Emmanouil Benetos, Xiang Yue, Chenghua Lin, Xu Tan, Stephen W. Huang, Jie Fu, and Ge Zhang. Mupt: A generative symbolic music pretrained transformer, 2024. 1

[5] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision, 2021. 1, 3

[6] Y. Wang, S. Wu, J. Hu, et al. Notagen: Advancing musicality in symbolic music generation with large language model training paradigms. *arXiv preprint arXiv:2502.18008*, 2025. 2